

Derek Serrano

www.derekserrano.dev • (416) 454-4057 • derek.ioab@icloud.com • <https://github.com/derekserr> • [linkedin.com/in/derek-serrano](https://www.linkedin.com/in/derek-serrano)

Backend Software Engineer with experience building scalable, distributed systems and customer-facing platforms using TypeScript/Node.js and Java.

EXPERIENCE

Software Engineer – Intact Insurance June 2024 – present

- Built and shipped customer-facing SaaS features using TypeScript and Java across a distributed architecture of 20+ microservices serving 50K+ policyholders
- Owned development and production support for 3–5 backend services, collaborating with cross-functional teams to deliver bi-weekly releases for \$1M insurance products
- Improved API performance by 30% by implementing Redis caching and Kafka-based asynchronous messaging across Kubernetes-hosted microservices
- Reduced MTTR by 35% and maintained 99.9% uptime by improving observability, alerting, and production diagnostics across distributed services
- Led migration of 10+ services from Spring Framework 5 to 6, resolving breaking changes and preventing production regressions during rollout
- Designed scalable REST APIs and optimized PostgreSQL/MongoDB schemas for distributed services, improving integration reliability and query performance

Software Engineer Intern – Intact Insurance May 2022 – Aug 2023

- Contributed to backend service development and bug fixes within a Java-based microservices system
- Gained hands-on experience with CI/CD pipelines, code reviews, and production release processes in distributed systems

Machine Learning Research Assistant – Healthcare Systems R&A May 2021 – August 2021

- Collaborated with researchers and engineers to develop vision-based image processing tools for lab testing workflows
- Implemented and evaluated machine learning models including SVMs and neural networks, contributing to data preprocessing, model validation, and performance evaluation in experimental pipelines

TECHNICAL SKILLS

Languages: Java, Python, TypeScript, JavaScript, SQL

Frontend: React, Angular

Backend: Node.js, GraphQL, REST APIs, SpringBoot, Microservices

Infrastructure & Cloud: Kubernetes, Docker, AWS, OpenShift, Kafka, Redis

Databases: Oracle, PostgreSQL, MongoDB

Tools: Jenkins, Git, Elasticsearch, Dynatrace

EDUCATION

Bachelor of Computer Engineering / McMaster University • 2019-2024 • GPA: 3.8